Banking under Conflict: Managers and Organizational Design*

Job Market Paper

Nicola Limodio[†] **Luca Picariello**[‡] **Tom Schwantje**[§]

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Abstract

How do organizations adapt internally when ethnic divisions intensify? We develop a model in which organizations jointly choose managerial appointments and delegation when locally matched managers have better information but are less aligned with headquarters, and test it using a panel of Ethiopian bank branches. Exploiting variation in banks' exposure to ethnic conflict across their branch networks, we find that conflict increases the appointment of locally matched managers, while reducing those managers' lending autonomy. Conflict-exposed branches are more likely to be staffed by insiders reassigned within the bank. An LLM-based CEO vignette exercise corroborates this mechanism.

JEL Codes: D23, D74, G21, O12.

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[†] **Bocconi University, Dept. of Finance, BAFFI and IGIER, and CEPR**: nicola.limodio@unibocconi.it

[‡] **Univ. of Naples Federico II, Dept. of Econ. & Stat., CSEF, MoFiR**: luca.picariello@unina.it

[§] **Bocconi University, Dept. of Finance, BAFFI and IGIER**: tom.schwantje@unibocconi.it

1 Introduction

How do organizations allocate decision-making authority when they operate across socially divided populations, and how do they adapt when those divisions deepen? Organizations face a well-known trade-off between local knowledge and internal communication costs (Garicano, 2000): delegating decisions to better-informed local agents improves the use of specialized knowledge but exposes the organization to those agents' biases (Dessein, 2002; Alonso, Dessein and Matouschek, 2008). In socially divided environments, this trade-off takes on an additional dimension, as employees' group identity can shape organizational design. Matching employees to customers by group identity may improve access to local information, but also create incentives for employees to favor those customers at the organization's expense. We examine these issues in the context of the banking sector – an ideal setting given its reliance on information and its comparative advantage in gathering and processing data to screen and monitor customers (Hertzberg, Liberti and Paravisini, 2010; Liberti and Mian, 2009).

Over the last decade, we have collected a novel panel dataset on the ethnicity of the manager and the organizational structure of bank branches in Ethiopia to study this issue. We also collect data highlighting a distinctive feature of banking in Ethiopia: banks are strongly associated with specific ethnic groups, as reflected in their leadership and dominant language (Regasa, Fielding and Roberts, 2022). Banks operating in areas populated by other ethnic groups must choose between hiring managers of the city's predominant ethnic group ("city-coethnic"), who are better informed about local borrowers, and less-informed managers coethnic with the bank ("bank-coethnic"), who facilitate better communication about those clients within the bank.

The empirical identification of how banks respond to changes in this information-alignment trade-off is challenging, as it requires variation exogenous to both bank-level and local conditions. To address this, we focus on a source of variation that affects the costs of information acquisition and communication within banks, without directly influencing local conditions. Specifically, we study the effect of banks' exposure to conflict involving the bank's ethnic group. To separate this effect from local economic conditions, we measure exposure through a bank's branch network, excluding conflict events near the branch's own city (Hjort, 2014; Rohner, Thoenig and Zilibotti, 2013; Korovkin and Makarin, 2023).[1] Using a panel of 979 branches operated by 16 banks before and after the onset of the 2020 civil war, we estimate how this network-based exposure reshapes branch organization and lending, controlling for branch and time fixed effects.

To guide our analysis, we develop a model of managerial assignment and delegation building on Dessein (2002) and Alonso, Dessein and Matouschek (2008). The key extension relative

[1] We use data from the Armed Conflict Location and Event Data Project (ACLED) to measure the location, timing, actors and intensity of conflict.

to these frameworks is that the bank jointly chooses the ethnicity of the branch manager and the authority delegated to that manager. City-coethnic managers may be better able to acquire soft information, but may also be less aligned with headquarters and more prone to in-group favoritism. When ethnic conflict intensifies, this trade-off between information and alignment sharpens. The model yields an unambiguous prediction for delegation: headquarters reduces the authority delegated to managers who match the city's ethnicity. The prediction for manager ethnicity, however, is ambiguous – the optimal adjustment depends on the relative effects of conflict on information acquisition and incentive alignment.

We first document a sharp shift in manager ethnicity in cities where the predominant local ethnicity differs from the bank's dominant ethnic group. In 2018, before the civil war, 77 percent of branch managers in these cities were coethnic with the bank; by 2022, that share had fallen to 35 percent. We then show that this shift is concentrated in branches more exposed, through the bank's branch network, to ethnic conflict involving the bank's ethnic group. Greater conflict exposure increases the likelihood that these branches appoint city-coethnic managers, while reducing those managers' lending autonomy, measured by the largest loan they can approve without headquarters' approval. This pattern suggests that banks respond to ethnic conflict by relying more on managers who share customers' ethnicity to acquire local information, while limiting the resulting incentive costs by lowering loan approval limits. Consistent with this interpretation, the greater the share of the population that shares the manager's ethnicity, the less authority the manager is granted, in particular after the onset of the conflict. Conflict-exposed branches are also more likely to be staffed by experienced managers reassigned from elsewhere in the same bank, suggesting greater reliance on trusted insiders. By contrast, we find no evidence of a change in the choice of manager ethnicity in bank-coethnic cities.

We then examine how network conflict exposure changes branch-level lending. In cities not coethnic with the bank, exposed branches issue fewer loans, but we find no evidence for an effect on the size of the geographic area these branches serve or average loan size. In bank-coethnic cities, exposed branches serve customers closer to the branch, and increase the average loan size. Lending rates, collateral requirements, and default rates appear unaffected. This suggests lending activity is sustained through these organizational adjustments.

We complement the main analysis with a vignette exercise using LLM-generated synthetic bank CEOs, following Horton, Filippas and Manning (2023). Across branch-year scenarios constructed from our survey data and bank annual reports, we vary the ethnicity of the hypothetical branch manager and the branch network's exposure to ethnic conflict. The synthetic CEOs mirror the same trade-off highlighted by the model and the data: they view managers who match the city's population as better able to acquire local information, but also as more prone to favoritism and harder to align with headquarters, leading to tighter delegation when conflict exposure is high.

We use this exercise to discipline interpretation of our findings and to open the organizational black box.

We corroborate the main findings using an instrument based on banks' pre-conflict proximity to cities historically associated with Beta Israel communities. The migration of Ethiopian Jews from these areas is linked to later disputed border configurations, which in turn predict subsequent exposure to ethnic conflict through the bank's branch network. The resulting estimates support our main organizational and lending results. Finally, we show that the combination of the patterns we document is difficult to reconcile with alternative mechanisms, including managers' own safety concerns, customers' preferences for coethnic managers, and employees' preferences for managers who share their ethnicity.

We contribute to three bodies of literature. First, we contribute to the literature on conflict by showing how violent ethnic conflict reshapes the internal organization of firms. Recent work shows that conflict can reorganize economic and social activity well beyond directly exposed markets (Fetzer et al., 2021; Sviatschi, 2022; Melnikov, Schmidt-Padilla and Sviatschi, 2025; Korovkin, Makarin and Miyauchi, 2025), and distort investment and other economic interactions (Fisman et al., 2020; Hjort, Song and Yenkey, 2025). We show that firms also respond by reorganizing internally, changing both who holds authority and how much authority is delegated. We complement Hjort (2014), who studies firms' adjustment of a single incentive margin, by documenting a broader set of organizational responses that help sustain lending under ethnic conflict. Specifically in banking, our paper joins a growing body of evidence on both the costs and the benefits of matching customers and employees along group lines: Fisman, Paravisini and Vig (2017) and Frame et al. (2025) study the benefits of matching in lending, while D'Acunto, Ghosh and Rossi (2026), Fisman et al. (2020) and Eichengreen and Saka (2026) study how cultural proximity and stereotypes can distort credit allocation.

Second, we contribute to organizational economics by providing an empirical setting to study the trade-off between local information and incentive alignment that is central to theories of delegation (Aghion and Tirole, 1997; Dessein, 2002; Alonso, Dessein and Matouschek, 2008). The theoretical literature typically varies the allocation of authority while holding the agent fixed, while the assignment literature varies who is appointed while holding authority fixed (Xu, Bertrand and Burgess, 2023; Dlugosz et al., 2024). Our setting allows us to study how organizations jointly adjust these margins when external conditions shift. This highlights how organizations actively manage the distortions documented in a recent finance literature: social ties can improve the quality of credit, but cultural proximity and stereotypes can also distort lending decisions (Brock and De Haas, 2023; Rehbein and Rother, 2025).

Third, we contribute to the literature on diversity and economic performance, showing that in polarized environments firms may optimally increase customer-facing diversity while simultaneously tightening internal controls. This highlights an important caveat to a recent literature that emphasizes the value of in-group homogeneity within firms (e.g., Espinosa and Delfino, 2024; Spenkuch, Teso and Xu, 2023; Colonnelli, Neto and Teso, 2025; Benson, Board and Meyer-ter Vehn, 2024; Ghosh, 2025), and complements evidence that diversity within organizations can improve performance (Rasul and Rogger, 2015; Lu, Naik and Teo, 2024). These firm-level adaptations help connect the macroeconomic literature on the negative link between diversity and economic growth (Easterly and Levine, 1997; Alesina and La Ferrara, 2005; La Ferrara, 2003; Miguel and Gugerty, 2005; Ashraf and Galor, 2013) with the microeconomic literature on inter-group frictions in economic interactions described above. Our findings also speak to how private organizations, alongside public institutions (Carlitz et al., 2025), adapt to govern inter-group frictions in divided societies.

The remainder of the paper proceeds as follows. Section 2 presents the theoretical framework and outlines the institutional context. In Section 3, we describe the data, our empirical testing strategy for the theoretical predictions, and the quasi-experimental variation in conflict stemming from the Beta Israel migration. Section 4 then details the empirical model, the main results and the synthetic CEO methodology. Section 5 considers alternative mechanisms. Finally, we discuss and interpret our findings in Section 6.

2 Model

Consider a bank operating a network of branches in a diverse environment in which a city's local population and the bank may be associated with different ethnic groups. The bank now faces a choice between hiring a manager coethnic with the bank and a manager coethnic with the city. This choice matters as the latter has better access to information from loan applicants who share her ethnicity (Fisman, Paravisini and Vig, 2017), but may also have an incentive to give those loan applicants excessively favorable lending terms from the bank's perspective (Fisman et al., 2020). Here, we develop a theoretical model building on the cheap-talk delegation literature (Aghion and Tirole, 1997; Dessein, 2002; Alonso, Dessein and Matouschek, 2008; Stein, 2002; Liberti and Mian, 2009) to formalize this trade-off. This model yields a joint prediction on the choice of the ethnicity of the manager and the amount of delegated authority to give that manager conditional on her ethnicity.

We model conflict as affecting both the relative ability of managers of different ethnic groups to acquire information from a loan applicant, and their bias toward members of their own group. These assumptions follow from an extensive literature that shows identity shapes economic behavior (Akerlof and Kranton, 2000), and that conflict affects how individuals identify with their own ethnic group and their general trust (Rohner, Thoenig and Zilibotti, 2013). Extant literature shows that exposure to conflict affects the biases of loan officers toward their own group (Fisman et al., 2020) and the biases of professionals more generally (Shayo and Zussman, 2017).

Environment. A loan has verifiable size $L \in [\underline{L}, \bar{L}]$ and a mean-zero payoff-relevant state s (the optimal lending stance) with variance V. The party holding authority chooses a noncontractible lending stance y, and HQ incurs $L(y-s)^2$. Let B denote a manager coethnic with the bank, C one coethnic with the city. Then, let $\pi_i \equiv \mathbb{P}(g = i)$ be the share of borrowers coethnic with manager i; when the bank and city groups differ, $\pi_C > \pi_B$. Manager types have the same ability to acquire information from and bias toward in-group loan applicants; differences in their behavior arise through π_i. Manager i has payoff

(1)
$$u_i(y, s, g, L) = -L\left[y - (s + b(\gamma)\mathbf{1}\{g = i\})\right]^2,$$

where $b(\gamma) > 0$ is own-group bias; she is aligned with HQ on out-group loans.

When a branch manager evaluates a loan, the loan applicant generates a mean-zero recordable posterior assessment x_i, with variance $\chi^I(\gamma)$ for an in-group borrower and $\chi^O(\gamma)$ for an out-group borrower. Because x_i is a posterior mean, a larger χ represents a more informative assessment.

When the branch manager and loan applicant share an ethnic group, this additionally generates soft information: an assessment $t_i \sim U[-q(\gamma), q(\gamma)]$. Define

(2)
$$\Delta_\chi(\gamma) \equiv \chi^I(\gamma) - \chi^O(\gamma) \geq 0, \qquad \bar{\chi}_i(\gamma) \equiv \chi^O(\gamma) + \pi_i \Delta_\chi(\gamma),$$

(3)
$$I_i(\gamma) \equiv V - \bar{\chi}_i(\gamma) - \pi_i \frac{q(\gamma)^2}{3}.$$

Conflict satisfies

(4)
$$b'(\gamma) > 0, \qquad q'(\gamma) \leq 0, \qquad \chi^{O\prime}(\gamma) < \chi^{I\prime}(\gamma) \leq 0,$$

so $\Delta'_\chi(\gamma) > 0$: cross-group information deteriorates more quickly than in-group information. See A.2 for a Bayesian implementation.

Before information is realized, HQ appoints a manager and commits to an authority schedule based on manager type and loan size. Under delegation, the manager chooses y. Under review, HQ observes x_i, receives a cheap-talk report about t_i on own-group loans, chooses y, and pays $\kappa > 0$. $\kappa > 0$ captures a fixed cost the HQ incurs to process a loan application. We do not allow the authority schedule to condition on the loan applicants' ethnicity. See A.1–A.4 for the microfoundations.

Authority. Under delegation and centralization, respectively, the expected losses are

$$\ell_i^{\mathrm{del}}(L) = \left(I_i + \pi_i b^2\right) L, \tag{5}$$

$$\ell_i^{\mathrm{hq}}(L) = \left(I_i + \pi_i \frac{q^2}{3}\right) L + \kappa. \tag{6}$$

Centralization removes the manager's bias, at the cost of soft information. Define

$$d(\gamma) \equiv b(\gamma)^2 - \frac{q(\gamma)^2}{3}, \qquad h_i(\gamma) \equiv \pi_i d(\gamma). \tag{7}$$

HQ delegates if and only if $h_i(\gamma)L \leq \kappa$. If $h_i \leq 0$, all loans are delegated; if $h_i > 0$, the approval limit is

$$\widehat{L}_i^*(\gamma) = \frac{\kappa}{h_i(\gamma)}. \tag{8}$$

Appointment. Let loan size have a continuous distribution F and mean $\mu \equiv \mathbb{E}[L]$. Anticipating the authority rule, HQ's operating loss under manager i is

$$\Lambda_i(\gamma) = V\mu - \bar{\chi}_i(\gamma)\mu + \mathbb{E}[\min\{\pi_i d(\gamma)L, \kappa\}]\,. \tag{9}$$

Let $w_i(\gamma)$ represent the ethnicity-specific assignment costs, varying with ethnicity and conflict due to potential safety or relocation costs, and set $\Omega_i \equiv \Lambda_i + w_i$. HQ chooses the candidate with the lowest Ω_i. For B and C, let $\Phi \equiv \Lambda_B - \Lambda_C$ and $\Psi \equiv \Omega_B - \Omega_C = \Phi + w_B - w_C$; conflict favors C when $\Psi' > 0$. See A.6 for the exact decomposition and A.7 for the assignment-cost extension.

2.1 Comparative statics and empirical interpretation

Let $D_i(L,\gamma) = \mathbf{1}\{\pi_i d(\gamma) L \leq \kappa\}$, $A_i = \mathbb{E}[D_i]$, and $M_{D,i} = \mathbb{E}[L D_i]$.

Proposition 1 (Appointment and authority)**.** *Suppose the primitives are differentiable, π_i is fixed in conflict, (4) holds, and F is continuous; for Part 1, suppose the bank and city groups differ, so $\pi_C > \pi_B$.*

(i). Conflict favors C if and only if

$$\mu(\pi_C - \pi_B)\Delta'_\chi + (w'_B - w'_C) > d'\left(\pi_C M_{D,C} - \pi_B M_{D,B}\right). \tag{10}$$

With equal assignment-cost responses, the interpretable comparison is

$$\Delta'_\chi(\gamma) > d'(\gamma), \tag{11}$$

which is sufficient in general and necessary and sufficient under full delegation.

(ii). Conflict weakly reduces authority conditional on manager type. For $\pi_i > 0$ and $\gamma_2 > \gamma_1$, $D_i(L, \gamma_2) \leq D_i(L, \gamma_1)$ for every L; whenever $h_i > 0$, $\widehat{L}_i^*$ strictly decreases.

(iii). Holding b and q fixed, a larger coethnic customer share weakly lowers authority. If $d > 0$, $\widehat{L}_i^* = \kappa/(\pi_i d)$ strictly decreases in $\pi_i > 0$; if $d \leq 0$, all loans remain delegated.

Derivation. The envelope theorem gives

$$\Psi' = \mu(\pi_C - \pi_B)\Delta'_\chi + (w'_B - w'_C) - d'(\pi_C M_{D,C} - \pi_B M_{D,B}).$$

Because $\pi_C > \pi_B$ implies $D_C \leq D_B$, $\pi_C M_{D,C} - \pi_B M_{D,B} \leq (\pi_C - \pi_B)\mu$, proving Part 1. Parts 2 and 3 follow from $d' = 2bb' - 2qq'/3 > 0$ and $h_i = \pi_i d$; aggregate authority changes strictly only when positive loan mass crosses the cutoff. □

Appointment identifies only whether the cross-group information premium and relative assignment costs dominate the own-group control wedge; it cannot separate these channels. The authority prediction is unambiguous: conflict weakly reduces authority within every manager type. Holding b and q fixed, managers with larger coethnic borrower shares have weakly less authority.

Corollary (appointment ordering). For any candidates j and k with $\pi_j > \pi_k$, condition (11) and equal assignment-cost responses imply

$$\frac{\mathrm{d}}{\mathrm{d}\gamma}(\Omega_j - \Omega_k) < 0. \tag{12}$$

Thus conflict favors candidates with larger local borrower shares. In particular, any third-group manager N with $\pi_N > \pi_B$ becomes more attractive relative to B; whether N is appointed depends on assignment costs and on π_N relative to the other candidates.

2.2 Banking in Ethiopia

The Ethiopian banking sector provides an ideal setting to study this mechanism for two reasons. First, Ethiopia is ethnically diverse, yet local areas are homogeneous. Second, banks have strong affiliations with specific ethnic groups, yet operate both in areas where the majority of the population shares their ethnicity and in cities where few people do.

In terms of ethnic diversity, Ethiopia is home to more than 90 ethnic groups, although four – the Oromo (34%), Amhara (30%), Somali (6%), and Tigray (6%) – account for roughly three-quarters of the population. Ethnicity is salient and discernible through appearance, language, and names (Abbay, 2004). The country's federal structure, established in 1992 after the fall of the military regime, organizes regions along ethnic lines. As a result, the major regions – including

Oromiya, Amhara, Somali, and Tigray – are highly homogeneous.[2] More broadly, over 85% of Ethiopians live in a woreda (district) where a single ethnic group constitutes at least 85% of the population. Appendix Figure E1 shows ethnic diversity at the local level.

Following liberalization in the early 1990s, privately owned banks entered the market, many with strong ties to specific ethnic groups and regions. These ethnic affiliations are evident in bank names, such as the "Cooperative Bank of Oromiya" and "Amhara Bank," branch locations (Regasa, Fielding and Roberts, 2022), the ethnicity of C-suite executives, and the predominant language used within banks. Despite these affiliations, banks now operate across ethnic boundaries. As competition intensified, banks established branches well beyond their home regions, often in cities where a different ethnic group predominates. This creates the trade-off our theoretical model formalizes: a bank can hire a city-coethnic manager to acquire better local information or a bank-coethnic manager to communicate more effectively with headquarters.

2.3 Conflict in Ethiopia

After relative peace and strong economic growth from 2000 to 2020, the Tigray War broke out in late 2020 following a rupture between the federal government and the Tigray People's Liberation Front (TPLF), the dominant political movement in Tigray. This setting suits our analysis for three reasons. First, conflict intensity varied sharply across time and space, with some areas experiencing severe violence while others remained relatively insulated. Second, violence involved multiple ethnic groups, reducing concerns that our estimates are driven by shocks specific to one group or region. Third, the conflict was unanticipated at the time of our 2018 data collection, supporting the exogeneity of pre-conflict bank and branch characteristics.

The war and subsequent civil conflict caused an estimated 300,000–800,000 civilian deaths by 2022 (Cedoca, 2024) and affected much of northern Ethiopia, with fighting concentrated in Tigray and along the north–south corridor linking Mekelle to Addis Ababa. Instability subsequently spread to Oromia (involving the Oromo Liberation Front, OLF) and to Amhara (involving Fano). Figure 1 summarizes conflict intensity over time (Panel A) and space (Panel B) in our data. Panel A shows that fatalities are positive but low and relatively stable during 2005–2019, then rise sharply starting in late 2020, with the number of fatalities peaking in 2021. Panel B maps events from 2019–2022 and indicates that conflict is widely distributed across central and northern Ethiopia, with substantial incidence in densely populated areas.[3]

The conflict would have been difficult to foresee when we collected our data in 2018. A

[2] Using microdata from the 2007 Ethiopian census (IPUMS), 96% of residents of Tigray are Tigrayan, 91% of residents of Amhara are Amhara, and 88% of residents of Oromiya are Oromo.

[3] Appendix Figure G2 shows the geographic distribution by year, with concentration in the north in 2020–2021 and new hotspots in other regions thereafter.

new prime minister had only just assumed power that year, beginning a political transition that culminated in the exclusion of the TPLF from the ruling coalition in late 2019. The exclusion left the TPLF with substantial military capacity but diminished political power, creating conditions conducive to conflict (Morelli, Ogliari and Hong, 2024). Because our data collection predates these developments, pre-conflict branch and manager attributes are unlikely to reflect anticipation of the violence. We exploit this timing in our empirical strategy.

3 Data and empirical strategy

3.1 Data

We use two main datasets for our analysis. The first is a novel branch-level panel covering the period before (2018) and after (2022) the onset of civil war in Ethiopia. The second is the Armed Conflict Location & Event Data Project (ACLED), which records the location, date, actors, and fatalities of conflict events. We complement this with several other sources detailed below.

Our initial sampling frame covered the universe of 3,232 Ethiopian bank branches at the time. We completed phone interviews with 1,910 branches in 2018 and targeted every one for follow-up in 2022. We re-interviewed 996 branches of 16 banks across 188 cities in 2022 to construct a panel. Our main analysis sample consists of the 979 branches that were interviewed in both waves and for which we could determine the location.[4] Appendix Table H2 shows that attrition is largely unrelated to our measure of conflict exposure, in particular in our main analysis sample. We therefore do not think attrition is likely to substantially bias our main results. Appendix F provides further detail on sample construction.

For each branch, we interview the most senior manager present at the time of the survey. The questionnaire covers three areas. First, we ask a series of questions about the manager's characteristics including their age, work experience, and education. Second, we administer a questionnaire we developed based on the World Management Survey (Bloom and Van Reenen, 2007), customized to reflect bank operations and lending activities, and ask questions about branch organization.[5] Finally, we ask a series of questions about lending at the branch, including the volume of lending, interest rates, collateral, non-performing loans, and the degree of autonomy the manager has to make lending decisions.

As ethnicity is highly salient in Ethiopia, we infer each manager's ethnic group based on the interview. To do so, two interviewers – the primary interviewer and an observer – jointly determine ethnicity based on the manager's region of birth, accent, spoken languages, and name.

[4] We exclude one re-interviewed branch because its identifier appears to link different branches across waves and 16 branches because of an issue encoding their locations.

[5] The management practices questions are the subject of a separate paper.

Appendix Table E3 reports the distribution of manager ethnicities for 2018 and 2022. For 261 managers in 2022, our enumerators did not record the ethnicity. For these managers, we infer their ethnicity based on (a) the manager's ethnicity in 2018 if they were already in the branch at the time (20 managers), or (b) the manager's place of birth (241 managers). Appendix Section I provides details on this imputation and shows that the main results related to ethnicity are robust to dropping branches where we imputed the manager's ethnicity.

Table 1 presents summary statistics for the main variables. Appendix Table F2 reports descriptive statistics for branches' organizational characteristics and lending activities by survey wave. Appendix F provides further details on survey design and implementation.

We use ACLED to measure the spatial intensity of ethnic conflict in Ethiopia. ACLED records the date, location (latitude/longitude), actors, and reported fatalities of events. To link events to ethnic groups, we use the actors listed in each event together with the Ethiopia Peace Observatory's actor–ethnicity classifications.[6] Throughout the paper, we keep all ACLED event types but restrict attention to events with at least one reported fatality. This focuses on high-severity incidents and reduces sensitivity to cross-area differences in the reporting of low-intensity episodes. Figure 1 summarizes this data, showing the sharp escalation of fatalities from ethnic conflict after 2020 (Panel A) and its spatial concentration during the civil-war period (Panel B).[7]

We also identify the ethnic affiliation of banks and cities. Ethiopian naming conventions – typically a first name followed by the father's first name, with no surnames – make it possible to infer ethnicity from publicly available records (Gebre, 2010; Fessha, 2016; Kefale, Kamusella and Van der Beken, 2021). We classify each bank by the ethnicity of its CEO and board members, assigning it to the ethnic group that holds the majority of board seats. We classify cities by their predominant ethnic group using the Murdock Ethnographic Atlas. Together with the manager-level data from our survey, this allows us to measure the ethnicity of banks, branch managers, and cities. Appendix E reports their distribution.[8]

Finally, we measure the ethnic composition of the population each branch serves using the IPUMS International 10-percent sample of the 2007 Ethiopian Population and Housing Census (Minnesota Population Center, 2020). We assign each respondent to the same ethnic groups we use for banks, managers, and cities, and compute population shares for each of Ethiopia's 714 woredas, each a third-level administrative unit. Each city takes the shares of the woreda containing

[6] The Ethiopia Peace Observatory is a special project run by ACLED. Appendix Table G1 reports the complete actor-to-ethnicity mapping.

[7] We use the ACLED Ethiopia extract downloaded on 15 May 2024 (and the Ethiopia Peace Observatory actor–ethnicity classifications accessed on 5 March 2025); we archive the extract used for analysis.

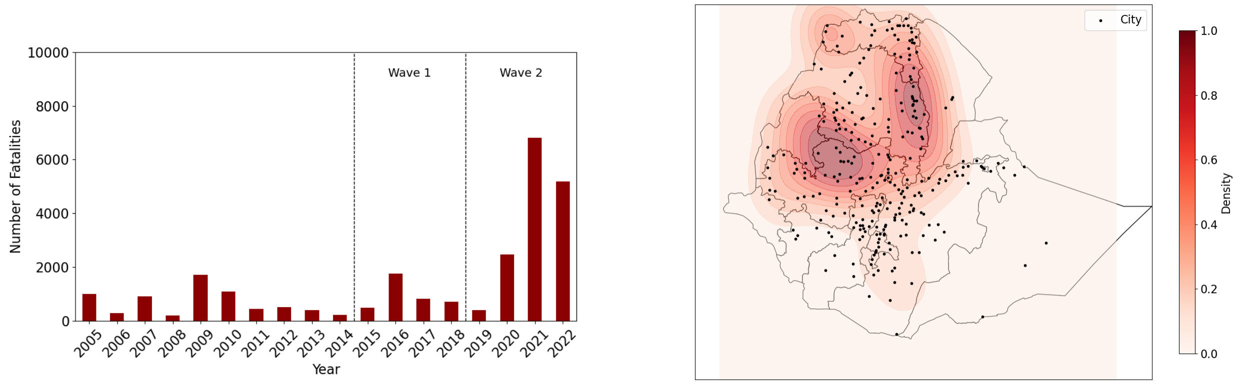
[8] We treat Harari as Oromo because the vast majority of Harari's population is Oromo. See Appendix E for details. We also aggregate ethnolinguistically distinct groups from Ethiopia's former SNNP region into a single category because they are individually too small in our sample to support separate classifications, and because they shared a common regional administrative structure. The results are robust to excluding this category.

Table 1: Summary statistics of analysis variables

	Panel A: Manager characteristics				
	N	Mean	Std. Dev.	Min	Max
Bank coethnic manager	1,957	0.65	0.48	0	1
City coethnic manager	1,957	0.54	0.50	0	1
Manager born in region	1,439	0.42	0.49	0	1
Tenure in bank (years)	1,954	7.71	4.44	0	36
Tenure in branch (years)	1,953	2.67	2.62	0	35
Age	1,896	34	5.67	16	60
University degree	1,956	0.99	0.07	0	1
Male manager	1,956	0.83	0.38	0	1
	Panel B: Branch characteristics				
	N	Mean	Std. Dev.	Min	Max
Share local employees	1,920	0.57	0.25	0.10	1.00
Branch employees	1,951	17.57	10.04	3	110
	Panel C: Lending outcomes				
	N	Mean	Std. Dev.	Min	Max
Operational distance (km)	1,700	6.80	4.85	0.05	50.00
Log number of loans	1,510	2.43	1.34	0.00	8.52
Log average loan size	1,505	6.30	3.13	-4.61	10.95
Average lending rate	1,637	0.15	0.02	0.10	0.20
Collateral (share of loan)	1,448	0.95	0.19	0.50	1.50
Share loan default	1,577	0.07	0.14	0.00	1.00

Notes: This table reports summary statistics for the main analysis variables based on the branch survey, organized into three panels. Panel A: Manager Characteristics. Bank-coethnic manager and city-coethnic manager indicate whether the manager shares the ethnicity of the bank or city, respectively. Manager born in region indicates whether the manager was born in the same region as the branch. Tenure in bank and branch are measured in years. University degree and male manager are dummy indicators. Panel B: Branch Characteristics. Share local employees is the share of branch employees from the local area. Branch employees is the number of employees in the branch. Panel C: Lending. Operational distance is the maximum distance (in km) at which a branch services customers. Log number of loans and log average loan size are in natural logarithms, with loan size expressed in ETB 1,000. Average lending rate is the mean rate on the past 10 loans. Collateral is the typical collateral as a share of total loan size. Share loan default is the share of defaulting loans in the past year.

Figure 1: Ethnic conflict over time and space



(a) Conflict over time

(b) Conflict over space

Notes: Panel A displays the number of fatalities from ethnic conflict in Ethiopia over time. The y-axis includes all fatalities from conflict involving at least one ethnic group in the ACLED database; this panel uses all events in ACLED. Panel B displays the lethality-weighted spatial intensity of ethnic conflict in Ethiopia from January 1, 2019 through December 31, 2022. Shaded areas are a kernel-density estimate of events weighted by fatalities and scaled to the colorbar. Black lines denote regional boundaries; black dots mark major cities.

it; Addis Ababa, administered as ten sub-city woredas, takes the population-weighted average of all ten.

3.2 Defining network-level exposure to conflict

We construct our main measure of exposure to ethnic conflict for each branch using ACLED. We define a branch's *network exposure* as the average exposure of the bank's branches outside the branch's own city to violence involving the bank's ethnic group. This measure captures ethnic conflict elsewhere in the bank's branch network while excluding exposure in the branch's own city.

Let $\mathcal{E}_t$ denote the set of ACLED conflict events in period t, and let f_e denote the number of fatalities in event $e \in \mathcal{E}_t$. For bank b, let $G_{be} \equiv \mathbf{1}\{e \text{ involves bank } b\text{'s ethnic group}\}$.[9]

Next, let $\mathcal{C}_b$ denote the set of cities in which bank b operated at least one branch in 2018 (its pre-conflict network). For each $c \in \mathcal{C}_b$, let n_{bc} denote the number of bank b's branches in city c, so that $N_b \equiv \sum_{c\in\mathcal{C}_b} n_{bc}$ is the total number of branches in its pre-conflict network. Let $c(i)$ denote the city in which branch i is located, and let $d(c, e)$ denote the distance, in kilometers, between city c and event e.

[9] That is, the ethnic group is mentioned as an actor in the conflict.

For branch i of bank b in period t, we define:

$$\text{NetworkExposure}_{ibt} = \frac{1}{N_b - n_{b,c(i)}}$$

$$\sum_{c \in \mathcal{C}_b} \underbrace{n_{bc}}_{\substack{\text{Branches of} \\ \text{bank } b \text{ in city } c}} \sum_{e \in \mathcal{E}_t} \underbrace{G_{be} f_e}_{\substack{\text{Fatalities from} \\ \text{conflict involving} \\ b\text{'s ethnic group}}} \times \underbrace{\mathbf{1}\{d(c,e) \leq 20\}}_{\substack{\text{Event } e \text{ occurs within} \\ \text{20 km of city } c}} \times \underbrace{\mathbf{1}\{d(c(i),e) > 20\}}_{\substack{\text{Event } e \text{ is not also within} \\ \text{20 km of the branch's} \\ \text{own city } c(i)}} \tag{13}$$

Intuitively, the measure captures the average number of fatalities from conflict events involving bank b's ethnic group that occur within 20 km of the cities in which the bank operates branches. We weight each city by the number of the bank's branches located there, exclude branch i's own city, and normalize by the number of branches the bank operates outside that city, $N_b - n_{b,c(i)}$. To match our two-period panel, we assign ACLED events occurring from 2015 through 2018 to period 1 and events occurring from 2019 through 2022 to period 2.[10]

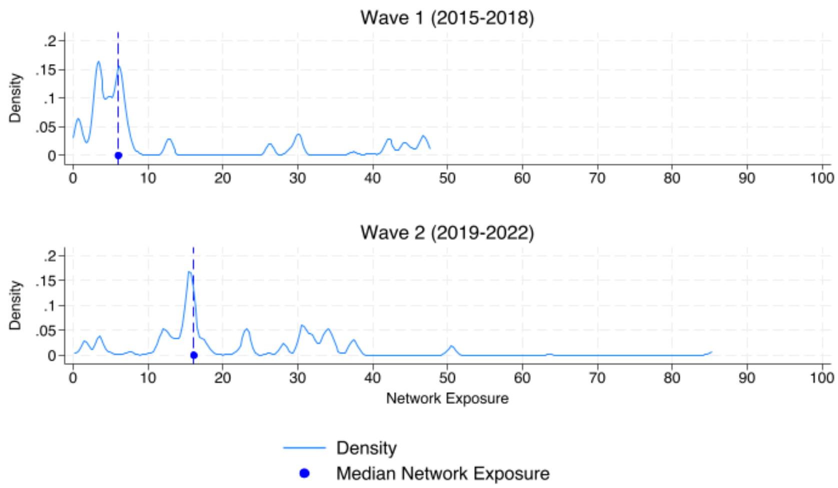
Figure 2 shows the distribution of branches' network exposure in the two periods covered by our panel: 2015–2018, leading up to the first survey, and 2019–2022, between the first and second surveys. Median exposure increased from 6.03 conflict-related fatalities per branch in 2015–2018 to 16.14 in 2019–2022. The upper tail also expanded substantially: the maximum rose from 47.7 fatalities per branch in the first period to more than 85 in the second. This highlights the sharp increase in banks' exposure to ethnic conflict between the first and second waves of our panel.

We interpret $\text{NetworkExposure}_{ibt}$ as the empirical counterpart of the conflict-salience parameter γ in Section 2: a shock to the ethnic identification and inter-group trust of bank b's employees and senior management, not to conditions in branch i's local market. The measure assumes that violence near a bank's branches, and so near its staff, is more salient to employees than violence far from the bank's branches. We therefore weight each city by the bank's branch presence there, so that $\text{NetworkExposure}_{ibt}$ is the average exposure to own-group violence across the bank's branches outside branch i's city; excluding i's own city keeps the measure from mechanically picking up violence in that local market. This measure thus captures how violence involving bank b's own ethnic group elsewhere in its network shifts ethnic identification and general trust among its personnel.

We use this network measure rather than a local one for two reasons. First, local conflict changes local economic conditions, and hence lending and investment, which can prompt branch reorganization for reasons unrelated to how information moves within the bank. Second, local conflict *involving a particular ethnic group* can work through three channels: it can (i) raise the salience of ethnicity for the manager, (ii) raise it for customers, or (iii) change the perceived safety

[10] In Appendix C.2, we examine robustness to the distance cutoff, alternative measures of conflict intensity—fatality counts and the number of fatal events—and the use of the 2022 rather than the 2018 branch network.

Figure 2: Distribution of network exposure over time



Notes: This figure displays the distribution of the strict-exclusion `NetworkExposure` measure from Equation 13 for two periods: 2015–2018 (top panel) and 2019–2022 (bottom panel). Each panel shows a kernel density (Gaussian, bandwidth 0.5) across our sample of branches and marks the period median. The median increases from 6.03 in 2015–2018 to 16.14 in 2019–2022.

of employees from that group. The network measure helps isolate the effect of ethnic conflict on the bank's personnel from its effects on customers and on local safety. It also builds on evidence that conflict disrupts inter-group economic exchange well beyond the areas where fighting occurs (Korovkin and Makarin, 2023).

To capture local conflict for branch i in city c, we define:

$$\text{LocalExposure}_{it} = \sum_{e \in \mathcal{E}_t} \underbrace{f_e}_{\text{Fatalities in event } e} \times \underbrace{\mathbf{1}\{d(c(i), e) < 20\}}_{\substack{\text{Event } e \text{ occurs within 20 km} \\ \text{of branch } i\text{'s city}}} . \tag{14}$$

We include this measure to more directly capture any potential direct local effects of conflict on the demand for loans.

3.3 Quasi-experimental variation in conflict

Much of the Amhara-Tigray conflict captured by our network exposure measures is concentrated along a stretch of the regional border between these regions, whose disputed status reflects a specific historical episode. The Beta Israel, commonly known as Ethiopian Jews, experienced significant migration to Israel during the 1980s and early 1990s. This movement was marked by notable operations such as Operation Moses in 1984 and Operation Solomon in 1991, during which tens of thousands were airlifted to Israel. This sudden shock plausibly resulted in contentious borders drawn in 1992, providing a separate source of conflict in the 2020 civil war. We use cities associated with this migration to construct an instrument for network-level exposure to conflict in subsection 4.5; here we outline the data and its link to modern-day conflict.

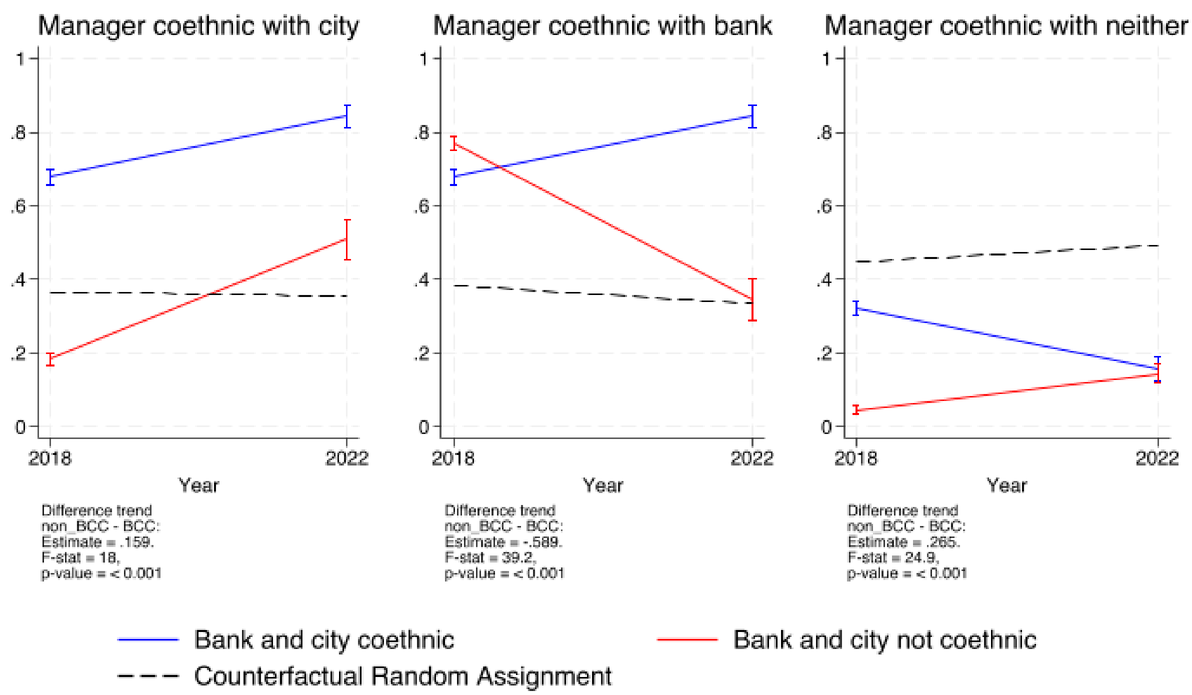
The most recent shifts of the Amhara-Tigray border prior to the 2020 conflict took place in 1992, when Ethiopia's ethnographic regional boundaries were drawn following the fall of the military regime. During these negotiations, dominated by the Tigray People's Liberation Front (TPLF), the border was drawn relatively far south toward Amhara, encompassing part of the area recently vacated by 40,000 Beta Israel (Nyssen and Demissie, 2023). In recent years, these disputed borders have continued to fuel tensions. The exclusion of the TPLF from Ethiopia's governing coalition in 2019 brought these long-standing grievances to the surface. The contested borders played a key role in the Amhara regional army's involvement in the Tigray conflict. The borders also fueled the continuing civil war, involving Amhara militias, who feared that territory they had gained might be returned to Tigray as part of a peace agreement (Nyssen and Demissie, 2023). These 1992 borders thus left disputed territories concentrated around the former Beta Israel settlements, which later became focal points of interethnic conflict. To study these cities, we extract data on the location of the main cities from which the Beta Israel departed (Kaplan, 1992). We then match these locations to banks and branches based on branch cities. We detail the exact empirical approach in Subsection 4.5.

3.4 Descriptive results

We begin by documenting how manager ethnicity changed from 2018 to 2022, separately for bank-coethnic and non-bank-coethnic cities. Using the branch-level panel observed in 2018 and 2022, we compare changes over time between branches located in cities that are coethnic with the bank and branches in cities that are not. Concretely, we estimate a fully interacted two-by-two specification with indicators for (i) the 2022 wave, (ii) whether the branch is in a non-bank-coethnic city, and (iii) their interaction; we report the implied group-by-year means in Figure 3. We run this analysis separately for two outcomes: (a) whether the manager is coethnic with the city and (b) whether the manager is coethnic with the bank.

Figure 3 shows that banks shifted from hiring bank-coethnic managers toward city-coethnic managers between 2018 and 2022. This shift is particularly pronounced in cities that are not coethnic with the bank. One interpretation, consistent with our framework, is that matching managers to local customers became more valuable relative to matching them to the bank; however, these descriptive patterns alone cannot distinguish that mechanism from alternative explanations. The empirical analysis that follows therefore focuses on identifying the causal effect of exposure to ethnic conflict on branch-level organization.

Figure 3: Changes in organizational structure over time



Notes: This figure presents the changes in organizational structure over time for two main outcome variables: an indicator for the manager and the bank being coethnic and an indicator for the manager and city being coethnic. This figure presents the changes in unconditional means for these outcomes, separately for branches operating in cities coethnic with the bank (bank-coethnic cities) and for branches operating in cities not coethnic with the bank (non-bank-coethnic cities). Finally, for the coethnicity variables we report the expected share of coethnic managers if the same sample of managers in terms of ethnic composition were to be randomly assigned to bank branches. We conduct this exercise 1,000 times and report the average of these simulations.

4 Empirical model and results

Our model makes two main predictions. First, ethnic conflict changes whether banks assign managers coethnic with the bank or with the city, but only in non-bank-coethnic cities—where the bank's ethnicity differs from the local population's. Cities that share the bank's ethnic group thus serve as a placebo. Second, ethnic conflict reduces the authority given to a manager when she shares the ethnicity of local customers. We therefore expect conflict to reduce authority most for managers who share the ethnicity of a larger share of local customers.

We first show that conflict exposure shifts branches in non-bank-coethnic cities from bank-coethnic managers to city-coethnic managers. We then examine two complementary organizational adjustments: reducing managers' lending authority and reassigning experienced insiders. Next, we show that branches in non-bank-coethnic cities make relatively limited adjustments to lending. We use a synthetic bank CEO to investigate the underlying mechanisms. Finally, we validate the main results using banks' proximity to Beta Israel cities as an instrument.

In our main specification, we estimate a two-way fixed-effects model to examine how within-branch changes in exposure to ethnic conflict through the bank's branch network affect branch-level outcomes.[11] We observe periods $t \in \{1, 2\}$ for branches $i \in \{1, \ldots, N\}$. These branches are in cities $c \in \{1, \ldots, C\}$, and part of banks $b \in \{1, \ldots, B\}$. We estimate the following regression specification:

$$Y_{ibtc} = \beta_1 \log(\text{NetworkExposure}_{ibt}) + \phi_i + \lambda_t + \varepsilon_{ibtc}, \tag{15}$$

where Y_{ibtc} is a branch-level outcome such as manager ethnicity or delegation. We include branch (ϕ_i) and time (λ_t) fixed effects and cluster standard errors at the branch level.[12]

This specification identifies the causal effect of network exposure to ethnic conflict under two conditions. First, changes in a branch's network exposure are plausibly exogenous if the spatial intensity of conflict was not predictable in 2018 and banks did not locate branches in anticipation of future conflict. We construct the bank's branch network using all branches interviewed in 2018, so post-2018 branch closures do not mechanically affect measured exposure. To further test this assumption, we implement an alternative design that uses the proximity of banks' pre-2018 branch networks to Beta Israel cities as a source of quasi-exogenous variation in exposure that was difficult

[11] Because we have two periods, this is equivalent to a first-difference specification, avoiding concerns about negative weights in staggered-adoption settings (Callaway and Sant'Anna, 2021).

[12] We add 0.01 to NetworkExposure because the measure is zero for a single branch; dropping that branch has virtually no effect on the estimates. We have also implemented wild-bootstrap inference (Cameron, Gelbach and Miller, 2008) at the bank level for the analyses in this section, reported in Appendix D. Under this more conservative approach the manager ethnicity results and the effect on the number of loans remain significant at the 5% level, while delegation and the manager reallocation and tenure effects are significant only at the 10% level.

to foresee in 2018. Second, Equation 15 identifies β_1 if no other time-varying factors jointly drive network exposure and branch organization. Branch fixed effects absorb time-invariant differences across branches, so identification requires that remaining shocks are common across branches (captured by time fixed effects) or otherwise orthogonal to changes in network exposure. A key threat is differential local trends or shocks in the cities where highly exposed branches operate. To address this, we re-estimate our main specification with city×time fixed effects and with direct controls for local conflict exposure.

To test formally whether bank-level exposure to conflict affects branches in bank-coethnic and non-bank-coethnic cities differently, we pool both groups and estimate the following interaction specification:

$$\begin{aligned} Y_{itbc} = {} & \beta_1 \ \log(\texttt{NetworkExposure})_{itb} + \\ & \beta_2 \ \log(\texttt{NetworkExposure})_{itb} \cdot \texttt{I[non\_bank\_city\_coethnic]}_{bc} + \\ & \phi_i \ + \ \lambda_{t,BankCityCoethnic} \ + \ \varepsilon_{itbc}, \end{aligned} \tag{16}$$

The bank-coethnic × time fixed effects ($\lambda_{t,BankCityCoethnic}$) allow separate time shocks for branches in bank-coethnic and non-bank-coethnic cities, so β_1 captures the effect of conflict in bank-coethnic cities alone. Throughout the results section, we report the p-value for $\beta_2 = 0$: the null that conflict exposure affects branches in bank-coethnic and non-bank-coethnic cities equally.

4.1 Ethnic Conflict and Manager Ethnicity

Figure 3 showed that between 2018 and 2022, banks shifted from hiring managers coethnic with the bank to hiring managers coethnic with the city. To test whether this shift is driven by exposure to violent ethnic conflict through the bank's branch network, we estimate equation 15 using indicators for whether the manager is coethnic with the bank or with the city as outcomes.

We find that network-level exposure to ethnic conflict shifts banks toward hiring city- rather than bank-coethnic managers, but *only* when the bank's and city's dominant ethnicities differ. As shown in Table 2, branches in cities not coethnic with the bank become 6.7 percentage points (p.p.) more likely to hire a city-coethnic manager, a 36.3% increase over the 2018 mean of 18.5%, in response to a one-standard-deviation increase in ethnic conflict exposure through the bank's branch network. Correspondingly, these branches become 13.0 p.p. less likely to hire a bank-coethnic manager, a 16.9% drop from the 2018 mean of 77.0%. The additional specifications in Table 2 show that this result is robust to either adding city-time fixed effects or controlling for local conflict intensity.[13]

[13] In Panel A, the bank- and city-coethnic outcomes coincide because the bank and city share an ethnicity.

Table 2: Branch manager ethnicity

	Panel A: Branches in Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-0.005	0.070	-0.006	-0.005	0.070	-0.006
	(0.026)	(0.055)	(0.026)	(0.026)	(0.055)	(0.026)
Local Exposure			0.014^{*}			0.014^{*}
			(0.008)			(0.008)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.679	0.679	0.679	0.679	0.679	0.679
N (Branches)	448	371	448	448	371	448

	Panel B: Branches in Non-Bank-Coethnic Cities					
	City coethnic manager			Bank coethnic manager		
Network Exposure	0.050^{***}	0.047^{***}	0.051^{***}	-0.097^{***}	-0.096^{***}	-0.099^{***}
	(0.016)	(0.017)	(0.016)	(0.015)	(0.017)	(0.015)
Local Exposure			-0.014			0.031^{***}
			(0.009)			(0.010)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	0.185	0.185	0.185	0.770	0.770	0.770
p-value	0.075	0.697	0.065	0.003	0.004	0.002
N (Branches)	531	464	531	531	464	531

Notes: The dependent variable is an indicator that equals 1 when a branch's manager is coethnic with the ethnic group associated with the city (columns 1–3) or with the ethnic group associated with the branch's bank (columns 4–6). Network exposure is the log of network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in equation 13. Local exposure is the log of conflict intensity near the city as defined in equation 14. The mean (SD) of the within-branch change in log network exposure from 2018 to 2022 is 0.953 (1.343). Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the test that the slope on network exposure is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

In contrast, for branches in bank-coethnic cities, the estimated effect of network exposure on the probability of employing a city-coethnic manager is statistically indistinguishable from zero. In the baseline specification, this estimate differs significantly from the corresponding estimate for branches in non-bank-coethnic cities. This difference is not robust to including city-by-time fixed effects: under that specification, the estimated effect in bank-coethnic cities is positive but statistically insignificant, and the difference between the two groups is also statistically insignificant. One possible interpretation is that this positive estimate captures the other side of the same bank-wide reassignment: bank-coethnic managers moved out of non-bank-coethnic cities may be reassigned to cities where their ethnicity matches both the bank and the local population. However, the evidence on reallocation in Table 4 is not consistent with this narrative as there is no increase in managers reallocated to these branches in bank-coethnic cities.

Finally, Appendix Table K1, Panel C, shows that network exposure to conflict also increases the share of branches whose manager is coethnic with neither the bank nor the city (+0.047, s.e. 0.012). With only 24 such branches in 2018 and 77 in 2022, our power to study this group is limited. Nonetheless, the same table provides suggestive evidence that these managers are appointed where the local population contains a large group outside both the bank's and the city's ethnic groups—i.e., where the information–incentive trade-off in our framework makes a manager from a third group a natural appointment. In branches with a neither-coethnic manager, the largest such group accounts for 28 percent of the city population – and the manager's own group for 26 percent – against 14–19 percent in other branches, while the city's group makes up only 50 percent of the population, against 57–65 percent elsewhere (Panel A). Consistent with this reading, redefining each city's ethnic group as the largest group in the 2007 census rather than the Murdock assignment shifts part of the neither-coethnic response into the city-coethnic response (Panel B), and the neither-coethnic response is concentrated in cities where such an outside group exceeds 30 percent of the population (0.115 against 0.033 elsewhere), where the city-coethnic response is correspondingly absent (Panel C). The rise in managers coethnic with neither the bank nor the city thus appears to be driven by cities with a large third group, from which these managers are largely drawn.

We make several assumptions to construct our measure of conflict. Appendix Figures C1 and C2 show that the results are robust to fixing the branch network at its 2018 or 2022 composition, using the number of fatal events instead of fatalities to weight conflict, varying the conflict-matching radius (15 or 25 km), excluding the state-owned banks (CBE and CBB), excluding Addis Ababa branches, and dropping branches of banks affiliated with the Tigray ethnic group. We implement this last restriction because the conflict may have affected Tigray-affiliated banks differently given the Tigray ethnic group's distinctive role in it.

4.2 Organizational adjustments to manager ethnicity

Banks thus shift toward appointing city-coethnic branch managers in response to ethnic conflict. In the model, this implies that conflict raises the organizational loss under a bank-coethnic manager relative to a city-coethnic manager: a bank-coethnic manager's ability to acquire information about borrowers falls, or her reservation wage rises, faster than the incentive costs of appointing a city-coethnic manager grow. If this information–incentive trade-off drives our results, banks should make two further adjustments. First, they should reduce the lending autonomy they grant these managers. Second—beyond the model—we expect banks to fill the posts by reassigning out-group managers with whom they already have a long institutional relationship, since tenure inside the organization can ease the interethnic communication frictions that conflict creates (Ghosh, 2025).

Our theoretical model implies that managers coethnic with the city they work in receive less autonomy over lending decisions, and that this autonomy falls with the share of local customers who share the manager's ethnicity, π_i. We measure delegation in our survey as the largest loan a branch manager can approve without authorization from headquarters. We test this hypothesis in two steps. First, we estimate Equation (15) with delegation as the dependent variable. Table 3 reports the results. Panel A shows that in non-bank-coethnic cities, greater exposure to ethnic conflict reduces delegation. In bank-coethnic cities the estimated effect is similar in sign but statistically insignificant. This pattern could, however, reflect a general reduction in the autonomy banks grant their managers in response to conflict near the branch network, rather than the targeted response our model predicts.

To distinguish these explanations, we ask whether delegation differs systematically for managers who are coethnic with the city, and in particular for managers who share the ethnicity of a larger share of local customers. We regress delegation on indicators for whether the manager is coethnic with the city and coethnic with neither the bank nor the city, and on the share of the local population coethnic with the manager, which proxies for π_i. We include branch and time fixed effects. Were the reduction in lending authority a general response to uncertainty, it would not concentrate among these managers. Panel B of Table 3 shows that banks give less autonomy to city-coethnic managers, consistent with banks limiting the incentive costs of employing them, and also to managers coethnic with neither the city nor the bank, though this second estimate is not significant. The result is driven mainly by branches where a larger share of the local population shares the manager's ethnicity, and it appears in bank-coethnic and non-bank-coethnic cities alike—as the model implies, since a manager's incentive to favor local borrowers depends on how many of them share her ethnicity, not on whether the city matches the bank. The reduction in delegation is therefore concentrated among the managers most exposed to the bias our model describes, rather than spread evenly across a bank's managers. In Appendix Table J1, we additionally show this effect

Table 3: Delegation

Panel A: Ethnic conflict and delegation						
	Bank and city not coethnic			Bank and city coethnic		
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-633.1**	-666.1**	-627.0**	-111.9	-1178.3	-115.6
	(282.0)	(330.3)	(281.5)	(381.9)	(839.1)	(384.5)
Local Exposure			-60.7			-422.3**
			(135.6)			(172.1)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	No	Yes	Yes	No	Yes
City-Time FE	No	Yes	No	No	Yes	No
2018 Mean dep. var	1684.8	1721.8	1684.8	1852.8	1993.8	1852.8
p-value	0.272	0.568	0.283	0.272	0.568	0.283
N (Branches)	322	266	322	231	160	231

Panel B: Manager ethnicity and delegation						
	Bank and city not coethnic			Bank and city coethnic		
City coethnic manager	-1893.0**	-108.0		-3196.2***	-794.9	
	(815.3)	(1088.1)		(1007.0)	(1524.2)	
Coethnic with neither	-1556.2	-1367.5				
	(1242.3)	(1270.2)				
Share local population		-4363.1***	-4223.3***		-4529.2**	-5428.3***
coethnic with manager		(1453.9)	(1088.9)		(2045.9)	(1340.0)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	1684.8	1684.8	1684.8	1843.5	1843.5	1843.5
N (Branches)	322	322	322	230	230	230

Notes: The dependent variable in both panels is the largest loan a branch manager can approve without authorization from headquarters, measured in thousands of 2018 Ethiopian birr (ETB). Values reported in 2022 are deflated to 2018 prices using annual CPI inflation from the World Bank's World Development Indicators (indicator FP.CPI.TOTL.ZG). The dependent variable is winsorized at the 95th percentile within survey-period $\times$ bank-city-coethnicity cells. Panel A estimates equation 15 separately for branches in cities that are not coethnic with the bank (columns 1–3) and branches in bank-coethnic cities (columns 4–6). Columns 1 and 4 report the baseline specification, columns 2 and 5 replace time fixed effects with city-by-time fixed effects, and columns 3 and 6 add Local Exposure to the baseline specification. Network Exposure is defined in equation 13; Local Exposure is defined in equation 14. Panel B uses the full 2018–2022 branch panel and includes branch and time fixed effects throughout. Columns 1–3 restrict the sample to branches in cities that are not coethnic with the bank; columns 4–6 restrict it to branches in bank-coethnic cities. Columns 1 and 4 include manager-ethnicity indicators, columns 2 and 5 additionally include the share of the local population coethnic with the manager, and columns 3 and 6 include only this population share. City-coethnic manager equals one when the manager shares the city's ethnicity; coethnic with neither equals one when the manager shares neither the bank's nor the city's ethnicity. In columns 1 and 2, bank-coethnic managers are the omitted category. In columns 4 and 5, the bank's and city's ethnicities coincide, so the omitted category is managers coethnic with neither. Standard errors, clustered at the branch level, are reported in parentheses. The p-values reported in Panel A test equality of the Network Exposure coefficients across the two subsamples using equation 16. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

is more pronounced in the second period, when conflict intensity is higher, suggesting that banks tighten delegation as the incentive issue gets larger in line with the theoretical model.

An alternative explanation of this result is that city-coethnic managers simply always have less autonomy, and that the average fall in autonomy of managers due to network exposure to conflict is due to the switch from bank- to city-coethnic managers. As a purely descriptive test, we restrict the sample to 124 branches in cities not coethnic with the bank where the manager type does not change from 2018 to 2022. Appendix Table B2 shows that manager autonomy also fell in response to ethnic conflict in this subsample. The estimated reduction in delegation is slightly larger (-811.1), though less precisely estimated ($p = 0.136$). The manager-type comparisons use 120 branches after excluding four branches with a stable neither-coethnic manager. Across these branches, the authority given to city-coethnic managers rose by less than half as much as that given to bank-coethnic managers. Similarly, replacing the type indicator with the local population share coethnic with the manager shows that limits increased less for managers whose ethnic group accounts for a larger share of the local population. This suggests that changes in autonomy in response to ethnic conflict are not purely an artifact of changes in the composition of managers at these branches.

Our baseline network-exposure measure excludes conflict near the focal branch and therefore varies across branches within a bank and period. To include bank-time fixed effects, we instead use exposure over the bank's complete branch network, which is constant within bank-time cells, and identify its differential effect on non-bank-coethnic versus bank-coethnic cities; the exposure main effect is absorbed by the bank-time fixed effect. Appendix Table B1 shows that, relative to branches in bank-coethnic cities, network exposure to conflict causes banks in non-bank-coethnic cities to shift toward city-coethnic managers with less autonomy.

The second organizational response concerns the type of managers banks select. Specifically, we ask whether they are reallocated within the bank or hired since the outbreak of the conflict, and how their experience compares with other managers. Because conflict induces branches to appoint city-coethnic managers, understanding whether these managers are redeployed insiders or external hires speaks directly to whether the relevant mechanism behind these changes in managers' ethnicity relates primarily to the alignment of incentives or to safety concerns.

To study the source of new managers, we use the 2022 cross-section of our survey. We define four outcome variables from two underlying indicators: *(i)* a dummy indicating whether the manager joined the bank after the outbreak of conflict (in 2021 or 2022); *(ii)* a dummy for whether the manager joined the branch after the outbreak of the conflict (in 2021 or 2022); *(iii)* a *reallocated manager* indicator equal to one if the manager joined the branch after but the bank before the outbreak of the conflict; and *(iv)* a *new manager* indicator equal to one if the manager joined the bank, and thus the branch, after the outbreak of the conflict.[14] We then estimate the following cross-sectional model:

$$Y_{ibc} = \alpha + \beta_1 \log(\texttt{NetworkExposure}_{ib}) + \varepsilon_{ibc}. \tag{17}$$

We cluster standard errors at the branch level. We find that, in cities not coethnic with the bank, a one-SD increase in log network exposure is associated with an increase of about 10.1 percentage points in the probability of employing a reallocated manager, i.e., a manager who was in a different branch of the bank before 2021 (Table 4). In bank-coethnic cities, network exposure is instead correlated with a slightly higher probability of having a manager who is entirely new to the bank. This pattern is consistent with a strategy in which banks limit potential lending bias by assigning non-bank-coethnic managers with whom they have a pre-existing institutional relationship.

This result suggests that the bank-coethnic managers previously managing the branches in cities not coethnic with the bank are not being reallocated to manage branches in cities coethnic with the bank. This is inconsistent with the alternative reading of our appointment results: that managers are being withdrawn because of concerns about their (perceived) safety. Under that account, these managers would have to show up elsewhere in the branch network, which they do not: branches in bank-coethnic cities take on managers who are new to the bank rather than redeployed insiders.

Finally, we examine the characteristics of these managers. We construct a short manager profile including their tenure at the bank, tenure at the branch, age, education, and gender. Using our panel, we re-estimate Equation 15 for each characteristic separately for bank-coethnic and non-bank-coethnic cities. Table 5 shows that, in non-bank-coethnic cities, a one-standard-deviation increase in network exposure is associated with appointing managers who have 0.67 years more tenure at the bank and around 3.6 months more tenure at the branch, and who are on average 0.72 years older. We find an extremely small (0.3 p.p.), marginally significant negative effect of conflict on the probability that the appointed manager holds a university degree – presumably because 98.7% of managers hold university degrees – and a small, insignificant change in gender composition. Together, longer tenure and greater age describe a redeployed insider rather than an external hire, consistent with the reallocation results in Table 4.

Taken together, these results show that banks whose branches are exposed to conflict through the branch network appoint city-coethnic managers, but take two steps to address the associated incentive costs: they reduce the lending authority delegated to these managers, and they fill the posts by reassigning long-tenured insiders rather than hiring externally. These adjustments move together, and in the direction the model predicts: banks accept a manager whose local information

[14] We cannot identify promotion timing directly, since we only observe employment duration at the branch and bank, not the year of managerial appointment.

Table 4: Manager reallocation

Panel A: Bank and city coethnic				
	New manager bank	New manager branch	Reallocated employee	New hire
	(1)	(2)	(3)	(4)
Network Exposure	0.016*	-0.015	-0.027	0.011
	(0.009)	(0.032)	(0.031)	(0.008)
Mean dep. var	0.034	0.397	0.370	0.027
N (Branches)	447	446	446	446

Panel B: Bank and city not coethnic				
	New manager bank	New manager branch	Reallocated employee	New hire
	(1)	(2)	(3)	(4)
Network Exposure	-0.010	0.065***	0.075***	-0.010
	(0.008)	(0.024)	(0.024)	(0.008)
Mean dep. var	0.017	0.360	0.345	0.015
p-value	0.023	0.230	0.013	0.059
N (Branches)	528	528	528	528

Notes: This table uses only data collected in 2022. The four outcome variables indicate whether the manager (1) joined the bank in 2021 or 2022 (after the outbreak of conflict in November 2020), (2) joined the branch in 2021 or 2022, and, conditional on joining the branch in 2021 or 2022, (3) was reallocated from a different branch of the same bank or (4) joined the bank as a new employee and did not change branches. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in equation 13. Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 5: Manager characteristics

	Panel A: Bank and city coethnic				
	Tenure in bank	Tenure in branch	Age of manager	University degree	Male manager
	(1)	(2)	(3)	(4)	(5)
Network Exposure	0.029	-0.265**	-0.026	-0.002	0.008
	(0.230)	(0.126)	(0.354)	(0.002)	(0.028)
Branch FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	6.859	2.384	33.888	0.993	0.868
N	448	448	448	448	448

	Panel B: Bank and city not coethnic				
	Tenure in bank	Tenure in branch	Age of manager	University degree	Male manager
Network Exposure	0.498***	0.221**	0.538***	-0.002*	0.008
	(0.154)	(0.098)	(0.202)	(0.001)	(0.014)
Branch FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	6.839	2.362	33.542	0.987	0.864
p-value	0.214	0.009	0.019	0.579	0.438
N	531	531	531	531	531

Notes: This table focuses on five outcome variables: (1) the manager's tenure at the bank in years, (2) the manager's tenure at the branch in years, (3) the age of the manager in years, (4) whether the manager has a university degree (BA or MA), and (5) whether the manager is male. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in equation 13. Local exposure is the log of conflict intensity near the city as defined in equation 14. The mean (SD) of the within-branch change in log bank conflict intensity from 2018 to 2022 is 0.953 (1.343). Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value is for the t-test that the parameter estimates for branches in bank-coethnic cities (Panel A) and in non-bank-coethnic cities are the same. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

is valuable, then contain the bias that appointment introduces by reducing her autonomy and by choosing someone with a long relationship with the bank. A pure labor-supply account, in which managers avoid postings they consider unsafe, would not produce these adjustments—nor would it explain why bank-coethnic cities take on managers who are new to the bank rather than those displaced from elsewhere in the network. The results therefore highlight how banks actively manage the information–incentive trade-off our model predicts.

4.3 Branch-level lending activities

We next examine how network-level exposure to ethnic conflict affects branch-level lending and operations. Table 6 reports estimates from specification 15 for a range of outcomes related to credit supply and branch activities.

We begin with branches' operational radius, defined as the geographic area from which they attract customers. Branches in bank-coethnic cities experience a contraction in this radius, whereas there is no effect on branches in non-bank-coethnic cities. One interpretation, consistent with classic work on soft versus hard information in lending (Petersen and Rajan, 2002; Agarwal and Hauswald, 2010), is that conflict exposure pushes branches in bank-coethnic cities toward more relationship-based screening within local networks, concentrating activity among nearby borrowers. By contrast, this result suggests that in non-bank-coethnic cities, where branches increasingly appoint city-coethnic managers, the negative effect of conflict on soft information acquisition and the positive effect of a city-coethnic manager result in no total effect.

Turning to credit supply, branches in non-bank-coethnic cities issue fewer loans in response to conflict exposure, although average loan size does not change. This pattern is consistent with credit being concentrated among a smaller set of safer or better-known borrowers. In these branches, we do not detect meaningful changes in lending rates, collateral or default rates. In contrast, branches in bank-coethnic cities expand the average loan size without changing the number of loans or loan terms.

These lending patterns complement the organizational adjustments documented above. In non-bank-coethnic cities, the shift toward city-coethnic managers appears to ease customer-facing inter-ethnic information frictions, but not completely: branches still reduce lending on the extensive margin. In bank-coethnic cities, where customer-facing inter-ethnic frictions are less central, we interpret these bank-wide adjustments as responses to a more volatile environment, rather than changes driven by the ethnicity of local customers.

Table 6: Branch-level lending

Panel A: Bank and city coethnic						
	Operational distance	Log number of loans	Log average loan size	Average lending rate	Collateral percentage	Share loan default
	(1)	(2)	(3)	(4)	(5)	(6)
Network Exposure	-1.658**	0.049	0.395**	0.001	0.015	-0.026
	(0.681)	(0.117)	(0.158)	(0.001)	(0.013)	(0.017)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.739	2.184	6.063	0.146	1.010	0.050
N	308	224	222	292	202	273

Panel B: Bank and city not coethnic						
	Operational distance	Log number of loans	Log average loan size	Average lending rate	Collateral percentage	Share loan default
Network Exposure	-0.084	-0.233***	0.123	-0.000	-0.006	-0.007
	(0.157)	(0.051)	(0.133)	(0.001)	(0.008)	(0.007)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	7.518	1.929	6.042	0.148	0.998	0.046
p-value	0.000	0.013	0.308	0.906	0.057	0.179
N	413	307	304	366	267	325

Notes: The dependent variables are the branch's operational distance, i.e., how far from the branch its customers are located, the log of the number of loans the branch has outstanding, the log of the average size of outstanding loans, the lending rate on a typical loan, the percentage of collateral on a typical loan and the share of loans that default. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in equation 13. The mean (SD) of the within-branch change in log bank conflict intensity from 2018 to 2022 is 0.953 (1.343). Panel A restricts the sample to branches in bank-coethnic cities, while Panel B focuses on branches in non-bank-coethnic cities. The regressions include branch and time fixed effects. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

4.4 Exploring the mechanisms with Synthetic CEOs

Our model implies that the information-incentive trade-off, especially through delegation, is the central mechanism linking conflict to organizational adjustment. The reduced-form evidence is consistent with this, but cannot decompose the trade-off into its constituent channels. To do so, we run a vignette experiment with LLM-generated synthetic bank CEOs, varying (i) the ethnicity of the hypothetical branch manager and (ii) network-level conflict exposure. We use LLMs as synthetic economic agents (Horton, Filippas and Manning, 2023) to address two limitations of standard approaches. The mechanisms of interest, information acquisition and transmission, alignment with headquarters, and ethnic favoritism, are difficult to observe and measure directly. Relatedly, direct surveys of bank executives are poorly suited to eliciting sensitive beliefs about ethnicity. As Ludwig, Mullainathan and Rambachan (2025) note, LLMs can therefore be useful for structured mechanism exploration.[15]

Figure 4 summarizes the design; Appendix L provides full details. We construct vignettes from observed branch-year cells, each combining the bank's and city's ethnic affiliations, conflict exposure, and a GPT-5-generated summary of the bank's annual report (Bybee, 2025; Fernández-Fuertes, 2025). For each observed branch-year cell in 2018 and 2022, we compare a city-coethnic and a bank-coethnic manager under low and high network conflict (10th and 90th percentiles). We elicit the synthetic CEO's assessment of each manager along four dimensions: information acquisition, information transmission, coethnic favoritism, and safety concerns.[16] We then ask about their preferred organizational design (delegation, wages) and expected lending outcomes. We estimate LLM-implied conditional average treatment effects using the saturated specification:

$$
\begin{aligned}
y_{itv} = {} & \beta_1 \, \text{CityCoethnic}_{itv} + \beta_2 \, \text{HighConflict}_{itv} + \beta_3 \big(\text{CityCoethnic}_{itv} \times \text{HighConflict}_{itv}\big) \\
& + \phi_i + \lambda_t + \varepsilon_{itv},
\end{aligned} \tag{18}
$$

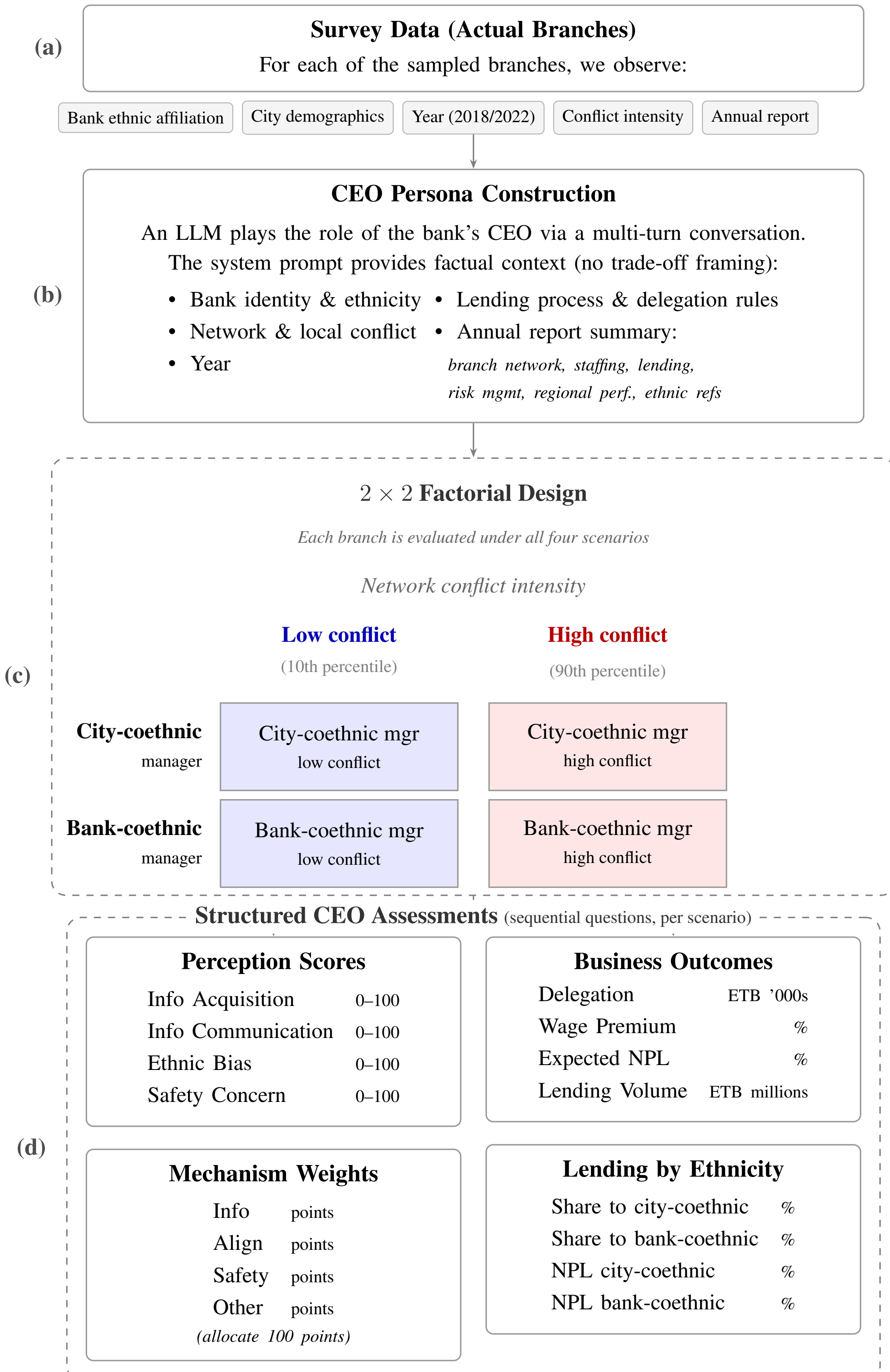
where i indexes branches, t indexes years, and v indexes vignette evaluations. CityCoethnic is an indicator for the manager and city being coethnic and $\text{HighConflict}_{itv}$ is the high-conflict treatment indicator minus its sample mean. Consequently, β_1 is the city-coethnic versus bank-coethnic contrast averaged across the two conflict conditions, β_2 is the high-minus-low effect for bank-coethnic managers, and β_3 is the differential high-minus-low effect for city-coethnic managers.

Table 7 reports the results. City-coethnic managers are perceived as better at acquiring local

[15] Importantly, the version of GPT-5 used in this exercise has a training cutoff that predates the first public circulation of this paper, so the model's responses cannot reflect exposure to our specific framework, hypotheses, or results.

[16] On the latter, we specifically ask about safety concerns, since a manager who is not coethnic with the local population may require a higher wage to induce them to live and work in that city when ethnic conflict intensifies, raising the bank's cost of assigning them there.

Figure 4: AI survey experiment design.



Notes: We generate a synthetic bank CEO and ask it to evaluate hypothetical managers under a 2 × 2 factorial design varying manager ethnicity (city-coethnic vs. bank-coethnic) and conflict intensity (10th vs. 90th percentile). For each scenario, the CEO provides structured assessments across eight outcome dimensions (panel d, top), mechanism weights, and expected lending patterns by customer ethnicity (panel d, bottom).

information but worse at transmitting it to headquarters, more prone to coethnic favoritism, and as having fewer safety concerns. These assessments translate into organizational choices: city-coethnic managers receive less delegation and lower wage premia, and are expected to generate higher lending volumes but worse loan quality. Higher conflict exposure weakens information acquisition for bank-coethnic managers and information transmission for city-coethnic managers, while increasing favoritism and safety concerns for both. Conflict reduces delegation and lending for both manager types, with a larger lending decline for city-coethnic managers. The vignette evidence thus decomposes the model's central trade-off into specific channels and mirrors the main reduced-form patterns in the data.

4.5 Robustness using exposure to Beta Israel Cities

To validate our baseline results, we exploit predetermined variation in banks' exposure to historically significant Beta Israel cities, which became disproportionately prone to ethnic conflict after the decline of Tigrayan influence in the federal government (Section 3). We define Z_{bt} as the share of bank b's 2018 branch network located within 100 km of a Beta Israel city, normalized to the $[0, 1]$ interval and interacted with an indicator for 2022. The estimates therefore compare the 2018–2022 change for branches of the banks with the greatest and least exposure; the interquartile range among non-bank-coethnic cities is 0.426. Because the measure is fixed by pre-conflict branch geography and tied to a historically specific component of the later Amhara–Tigray border dispute, it is unlikely to reflect endogenous branch relocation or other contemporaneous bank responses. Appendix Figure C3 maps the Beta Israel cities and the exposure measure.[17]

Table 8 presents the results for managers' ethnicity, delegation and other manager characteristics. Consistent with our baseline estimates, we find no significant effects in bank-coethnic cities, except a decrease in managers' branch tenure. In contrast, in non-bank-coethnic cities, variation in the share of branches near Beta Israel cities leads to a marked differential decline in the probability of appointing a bank-coethnic manager and a corresponding rise in the probability of appointing a city-coethnic manager. These results reinforce our main finding that banks increasingly rely on locally coethnic managers when exposed to higher conflict risk. Similarly, the estimates of the relative reduction in delegation are qualitatively similar, as are those for managers' bank tenure, across cities that are and are not coethnic with the bank.

Appendix Table C5 reports the corresponding results for branch-level lending.

[17] Appendix Table C6 shows that the results are robust to using a 50 km radius. We obtain similar results using only branches located in Beta Israel cities.

Table 7: Synthetic CEO Expectations of Effects of Conflict

Panel A: The effect of conflict on mechanisms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CEO Assessments				Mechanism Weights			
	Info Acq.	Info Comm.	Ethnic Bias	Safety Concern	Info	Align	Safety	Other
City-coethnic	54.52***	-22.83***	39.88***	-27.80***	12.47***	-7.01***	-9.19***	3.73***
	(0.35)	(0.47)	(0.68)	(0.81)	(0.31)	(0.33)	(0.30)	(0.24)
High conflict	-4.61***	-1.34***	5.63***	30.68***	-4.52***	-3.18***	9.11***	-1.41***
	(0.43)	(0.39)	(1.04)	(0.93)	(0.44)	(0.44)	(0.49)	(0.28)
High conflict	3.91***	-2.75***	-3.50***	-8.45***	1.37**	3.26***	-3.51***	-1.12***
× City-coeth	(0.54)	(0.87)	(1.08)	(1.47)	(0.63)	(0.58)	(0.57)	(0.42)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. var. mean	58.04	69.28	51.49	44.49	34.77	37.43	18.45	9.35
Observations	742	742	742	742	742	742	742	742
R^2	0.985	0.823	0.884	0.822	0.748	0.525	0.721	0.389

Panel B: The effect of conflict on outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Outcomes				Lending (%)		NPL (%)	
	Deleg.	Wage Prem.	Exp. NPL	Lend Vol.	City-Coeth.	Bank-Coeth.	City-Coeth.	Bank-Coeth.
City-coethnic	-158.3***	-25.73***	0.77***	44.30***	1.26***	-5.12***	1.91***	-2.69***
	(30.28)	(0.50)	(0.12)	(1.83)	(0.46)	(0.42)	(0.16)	(0.25)
High conflict	-174.2***	9.50***	2.05***	-11.76***	-2.37***	1.97***	1.70***	3.74***
	(40.25)	(0.37)	(0.16)	(1.37)	(0.61)	(0.54)	(0.16)	(0.39)
High conflict	-2.03	-2.40***	0.39*	-14.67***	3.25***	-2.45***	1.19***	-2.63***
× City-coeth	(48.78)	(0.88)	(0.21)	(2.93)	(0.76)	(0.62)	(0.28)	(0.40)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. var. mean	704.85	9.73	7.77	94.08	65.88	17.80	8.38	6.83
Observations	742	742	742	742	742	742	742	742
R^2	0.443	0.856	0.583	0.751	0.510	0.511	0.568	0.439

Notes: This table reports OLS estimates from an AI survey experiment in which a large language model (LLM) plays the role of a bank CEO evaluating hypothetical branch managers described in standardized vignettes. Each vignette varies (i) whether the manager is city-coethnic or bank-coethnic and (ii) whether the branch is assigned high or low conflict intensity. The specification includes indicators for City-coethnic, High conflict, and their interaction. City-coethnic equals one if the hypothetical manager's ethnicity matches the branch city's dominant ethnic group; bank-coethnic managers (ethnicity matching the bank headquarters' dominant ethnic group) are the omitted category. High conflict equals 0.5 for branches assigned conflict at the 90th percentile of the distribution of fatalities; it is -0.5 for low conflict (10th percentile), the omitted category. Panel A (Mechanisms). Columns (1)–(4) report LLM-elicited expectations on a 0–100 scale: Info Acq (ability to gather soft information about local borrowers), Info Comm (alignment of loan recommendations with headquarters' objectives), Ethnic Bias (likelihood of ethnic favoritism in lending), and Safety Concern (physical security risk). Columns (5)–(8) report the LLM's stated mechanism weights (shares in percentage points) assigned to Info, Align, Safety, and Other. Panel B (Outcomes). Columns (1)–(4) report Deleg. (maximum loan size approvable without headquarters authorization; ETB thousands), Wage Prem. (percent), Exp. NPL (percent), and Lend Vol. (ETB millions). Columns (5)–(6) report expected lending to city-coethnic and bank-coethnic borrowers (percent). Columns (7)–(8) report expected NPLs for city-coethnic and bank-coethnic borrowers (percent). The sample is restricted to branches where the bank headquarters' dominant ethnicity differs from the branch city's dominant ethnicity. All specifications include branch fixed effects and year fixed effects (as reported in the table). Standard errors clustered at the branch level are in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 8: Beta Israel variation and branch managers

	Panel A: Bank-City Coethnic					
	(1) City coethnic manager	(2) Bank coethnic manager	(3) Coethnic with neither	(4) Delegation	(5) Tenure in bank	(6) Tenure in branch
Beta Israel	0.050	0.050	-0.050	-1234	-1.455*	-1.253***
Exposure	(0.094)	(0.094)	(0.094)	(1978)	(0.814)	(0.416)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.680	0.680	0.320	1852.8	6.859	2.390
N	447	447	447	231	447	446

	Panel B: Not Bank-City Coethnic					
	City coethnic manager	Bank coethnic manager	Coethnic with neither	Delegation	Tenure in bank	Tenure in branch
Beta Israel	0.364***	-0.547***	0.183**	-6711***	1.007	0.408
Exposure	(0.112)	(0.122)	(0.075)	(1922)	(0.958)	(0.658)
Branch FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
2018 Mean dep. var	0.185	0.770	0.045	1684.8	6.837	2.366
p-value	0.032	0.000	0.053	0.047	0.050	0.033
N	531	531	531	322	528	528

Notes: The dependent variables are indicators that equal 1 when a branch's manager is coethnic with the ethnic group associated with the city (column 1), with the ethnic group associated with the branch's bank (column 2), or with neither group (column 3); the largest loan a branch manager can approve without headquarters' agreement (column 4), measured in thousands of 2018 ETB and winsorized at the 95th percentile within period $\times$ bank-city-coethnicity cells; and the manager's tenure in the bank (column 5) and branch (column 6), in years. In non-bank-coethnic cities, the three ethnicity categories in columns 1–3 are mutually exclusive and exhaustive, so their coefficients sum to zero by construction. Beta Israel Exposure is the normalized share of the bank's 2018 branch network located within 100 km of a historic Beta Israel city, interacted with a post-2018 indicator. Panel A restricts the sample to branches in bank-coethnic cities, while Panel B focuses on branches in non-bank-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported p-value tests whether the coefficient is identical across panels. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

5 Mechanism tests

In interpreting our empirical results, we have so far focused on the branch manager's role in the loan application process. However, branch managers perform multiple functions within the organization: they also supervise employees and attract customers to the branch. Moreover, ethnic conflict may directly affect a manager's preferences and reservation wage for living in cities with a different predominant ethnicity. We examine these three alternative channels in turn: (i) managers' own location preferences, (ii) customers' preferences for banks and managers that share their ethnicity, and (iii) frictions with local employees.

Managers' willingness to live in cities whose predominant ethnicity differs from their own could plausibly be affected by the intensity of ethnic conflict, whether due to local insecurity or a desire to return to one's hometown. If so, the changes in managerial composition we document would reflect individual relocation decisions rather than bank-level optimization. Ethiopian banks are highly hierarchical, and personnel assignments – including branch manager postings – are typically decided at headquarters with limited scope for voluntary transfers, making this interpretation less likely *a priori*. On the other hand, our measure of network-level exposure to conflict, though constructed to abstract from local conflict intensity, is positively correlated with it, making safety concerns a plausible alternative explanation.

We run three tests against this alternative. First, our main estimates (Table 2) are robust to city$\times$time fixed effects and to direct controls for local conflict intensity. If fear of local conflict were the primary driver, these controls should substantially attenuate the coefficients; instead, they hardly move them. Under these fixed effects, the comparison is between two branches in the same city, both non-coethnic with the local population, that differ only in their exposure to conflict involving their own ethnic group elsewhere in the network. Second, city-level controls cannot absorb threats directed at managers of the bank's ethnicity in particular. We therefore control directly for local conflict involving the bank's own ethnic group (Appendix Table J2). Our results are unchanged; if anything, higher local conflict involving the bank's ethnic group slightly raises the probability of appointing a bank-coethnic manager—the opposite sign from what personal security concerns would predict. Third, network-level conflict might heighten managers' perceived risk of being targeted even absent local threats. We therefore redefine exposure to include only events involving both the bank's and the city's ethnic groups (Appendix Figure J1). The estimated relationship attenuates: the city-coethnic coefficient is 0.039 against 0.049 in the main specification, and the bank-coethnic coefficient is not significant.

A second concern is that conflict erodes customers' trust in managers who do not share their ethnicity, leading them to take their business to a coethnic bank and banks to adjust staffing in response. If customers switch, the effect should be strongest where they have a ready alternative:

a branch of a bank coethnic with the local population. We test this by interacting conflict exposure with the presence of a city-coethnic competitor. Table J3 shows an interaction of 0.020 (s.e. 0.044), too small to account for more than 40% of the main effect. Results are similar when we measure competition by the number of branches in the city or by managers' perceived number of competitors in 2018.

Finally, branch managers not only communicate with headquarters but also supervise local employees. Conflict may therefore affect not only upward communication but also the manager's ability to motivate and coordinate subordinates. Li et al. (2025) formalize a version of this idea: in a three-layer hierarchy, a middle manager must be trusted by both the principal above and the agents below. A manager too aligned with the top does not motivate workers to exert effort; one too aligned with workers is discounted by headquarters. When conflict sharpens ethnic loyalties, a bank-coethnic manager may become too closely identified with headquarters, and a city-coethnic manager too closely identified with local staff. A manager coethnic with neither side may then be better positioned to sustain cooperation in both directions. If this channel were important, conflict should increase the probability that branches are run by managers coethnic with neither the bank nor the city, especially where the share of local employees is high.

We find support for the first part of this prediction. Appendix Table K1, Panel C, shows that a one-standard-deviation increase in conflict exposure raises the probability of appointing a neither-coethnic manager by 6.3 p.p., significant at the 1 percent level. Both accounts predict this shift, so the appointment result alone does not separate them. They differ on the delegation margin: the reduction in lending authority depends on borrowers' ethnic composition rather than staff composition. The second part of the prediction finds little support. The appointment effect is only slightly larger in branches with a higher baseline share of local employees, and the difference is not significant (Appendix Table J5). This implies that although this channel may be present, it cannot by itself generate our main results. The same panel shows that the rise in neither-coethnic managers is concentrated in cities with a large group coethnic with neither the bank nor the atlas-assigned city group, consistent with conflict reallocating appointments toward locally rooted groups rather than with a distinct supervision channel.

6 Discussion

This paper shows that ethnic conflict changes not only market outcomes but also the internal organization of firms. Using new branch-level data from Ethiopian banks, we find that greater conflict exposure leads branches in non-bank-coethnic cities to appoint managers who are coethnic with the city's predominant group rather than coethnic with the bank, reduce the authority delegated to those managers, and rely more on experienced insiders. The key implication is that firms are not passive in their response to shocks that intensify social divisions; they actively reorganize in order to keep operating across group lines.

Our results speak to a central question in organizational economics: how firms trade off local knowledge against incentive alignment (Aghion and Tirole, 1997; Dessein, 2002; Alonso, Dessein and Matouschek, 2008). In our setting, conflict raises the value of managers who can acquire soft information from local borrowers, but also sharpens the risk of biased lending and distorted communication with headquarters. Banks respond on both margins at once. They do not choose between local adaptation and central oversight; rather, they appoint managers better able to acquire local information while simultaneously tightening the control exercised by headquarters. This joint adjustment is difficult to reconcile with explanations in which managers relocate voluntarily or customers drive the change, since banks would then have little reason to simultaneously tighten delegation and reassign insiders.

More broadly, the findings suggest that identity divisions can reshape the production of information inside organizations. A growing literature studies how ethnicity, culture, or political affiliation affect lending and other economic interactions (Hjort, 2014; Fisman et al., 2020; Frame et al., 2025; D'Acunto, Ghosh and Rossi, 2026; Eichengreen and Saka, 2026). Our evidence highlights a distinct mechanism: social divisions affect who can acquire information, how credibly that information can be transmitted, and how authority should be allocated between headquarters and the branch. In that sense, firms not only passively suffer the effects of conflict, for example due to discrimination, but also actively manage those effects through their internal organization.

The Ethiopian banking sector provides a particularly sharp environment in which to study this mechanism, but the logic extends beyond this context. Many firms operate across salient identity boundaries, whether ethnic, linguistic, religious, or political. Whenever customer-facing information is locally embedded but organizations remain centrally governed, shocks that intensify identity salience can make local matching more valuable and delegation more costly. Our results therefore suggest a broader principle: firms may respond to polarization not by retreating from diverse markets, but by redesigning authority, monitoring, and personnel assignment to continue operating across group lines.

At the same time, our findings point to an important limit of adaptation. Our results suggest that the organizational changes we document help sustain lending performance: there is no evidence for deterioration in observed loan performance. This, however, does not mean they are costless. Tighter control can slow decision-making, reduce initiative (Aghion and Tirole, 1997), and require greater reliance on internal labor markets. More generally, the need to offset social frictions with additional oversight implies that polarization is costly even when firms successfully adjust. Understanding those longer-run costs, and whether similar responses arise in other sectors and political settings, is an important direction for future work.

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